

Database Management Systems — End Semester Examination (Sample) 2024

Part A — Short Answer Questions (2 marks each)

1. Define normalization and state why it is used in database design.
2. List the four ACID properties of a database transaction.
3. What is the difference between a primary key and a candidate key?
4. Define a B-tree index and state its purpose in query processing.

Part B — Long Answer Questions (10 marks each)

5. Explain First, Second, and Third Normal Forms with suitable examples for each.
6. Describe how concurrency control mechanisms preserve the isolation property of transactions.
7. Explain the role of query optimization in improving DBMS performance, with an example execution plan.